

# Standard Form & Indices

Test · 41 marks total · Sections F & G include Higher-tier content

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Score: \_\_\_\_\_ / 41

## Section A — Laws of indices

**A1.** Write  $4^5 \times 4^3$  as a single power of 4.

[1 mark]

**A2.** Write  $7^9 \div 7^2$  as a single power of 7.

[1 mark]

**A3.** Write  $(6^2)^3$  as a single power of 6.

[1 mark]

**A4.** Work out the value of  $2^3 \times 2^2$ .

[2 marks]

## Section B — Zero and negative indices

**B1.** Work out the value of  $12^0$ .

[1 mark]

**B2.** Work out the value of  $5^{-2}$ . Give your answer as a fraction.

[2 marks]

**B3.** Work out the value of  $10^{-3}$ . Give your answer as a decimal.

[2 marks]

## Section C — Writing standard form

**C1.** Write 72,000 in standard form.

[2 marks]

**C2.** Write 8,300,000 in standard form.

[2 marks]

**C3.** Write 0.0048 in standard form.

[2 marks]

**C4.** Write 0.00003 in standard form.

[2 marks]

## Section D — Converting back, and calculating

**D1.** Write  $6.5 \times 10^4$  as an ordinary number.

[1 mark]

**D2.** Write  $3 \times 10^{-5}$  as an ordinary number.

[1 mark]

**D3.** Work out  $(4 \times 10^3) \times (2 \times 10^5)$ . Give your answer in standard form.

[3 marks]

**D4.** Work out  $(9 \times 10^8) \div (3 \times 10^3)$ . Give your answer in standard form.

[3 marks]

## Section E — Powers to know, and estimating roots

**E1.** Write down the value of  $3^4$ .

[1 mark]

**E2.**  $\sqrt{60}$  lies between which two consecutive whole numbers?

[1 mark]

## Section F — Laws of indices with algebra

**F1.** Simplify  $a^6 \times a^4$ .

[1 mark]

**F2.** Simplify  $b^{10} \div b^3$ .

[1 mark]

**F3.** Simplify  $(c^3)^4$ .

[1 mark]

**F4.** Simplify  $5d^2 \times 2d^6$ .

[2 marks]

**F5.** Simplify  $(3e^2)^3$ .

[2 marks]

## Section G — Fractional indices (Higher)

**G1.** Work out the value of  $49^{1/2}$ .

[1 mark]

**G2.** Work out the value of  $27^{1/3}$ .

[1 mark]

**G3.** Work out the value of  $16^{3/4}$ .

[2 marks]

**G4.** Work out the value of  $9^{-1/2}$ .

[2 marks]